

Time discretisation, the monolithic approach

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Outline

- 1 Introduction
- 2 Time-integration schemes
 - Multi-step schemes
- 3 Stability and convergence of a time-integration scheme
 - Stiff ODEs
- 4 Space-time PDE discretization
 - Von Neumann's method
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In this lecture

In this lecture I will:

- Introduce the main time-discretization techniques
- Provide some tools to predict the stability and accuracy of a numerical scheme

A historical viewpoint

It must be said that:

- mainly due to memory requirements, the monolithic approach has been **seldom used** in the past
- it has been used only for 2D problems
- it is generally less efficient from the computational viewpoint

Why should we bother about a monolithic approach?

It is a preliminary step towards the approach we will use.

The discretised problem

Once the spatial discretisation has been carried out, we end up with a system of ordinary differential equations.

When the incompressible Navier–Stokes equations are discretised, this system is nonlinear. It can be written as

$$M \frac{dU}{dt} = F(U, t),$$

where

- U is the vector unknown (velocity and pressure),
- M is the mass matrix, for INSE it is singular
- F is a vector function of the unknowns and time (if forcing or b.c. are a function of time)

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Integrating an ODE

We start considering the problem of integrating a simple, first order ODE:

$$\frac{du}{dt} = f(u, t)$$

where we have a linear ODE if $f(u, t) = Au + g(t)$, nonlinear otherwise.

Theorem: existence and uniqueness

If f is Lipschitz continuous over the domain

$\mathcal{D} = \{(u, t) : |u - \eta| \leq a, t_0 \leq t \leq t_1\}$, i.e. $\exists L \geq 0$ so that

$|f(u, t) - f(u^*, t)| \leq L|u - u^*|$ in $\mathcal{D}$, then there is a unique solution up to time $T^* = \min(t_1, t_0 + a / \max_{(u,t) \in \mathcal{D}} |f(u, t)|)$

Taxonomy of time-integration schemes

There is a vast variety of time-integration schemes for ODEs. The most used numerical schemes for ODEs can be classified as follows:

- Single step methods, among which are
 - Taylor schemes
 - Runge–Kutta schemes
- **Linear multistep methods**, among which are
 - Adams–Bashforth methods
 - Adams–Moulton methods
 - Backward differentiation formulas (BDF)

Linear Multistep Methods

In linear multistep methods, several time steps are used to compute the solution at the new time step.

They are described by the following formula:

$$\sum_{j=0}^r \alpha_j U_{n+j} = \Delta t \sum_{j=0}^r \beta_j f(U_{n+j}, t_{n+j})$$

If $\beta_r = 0$, the method is explicit, otherwise it is implicit.

The coefficients are defined up to a multiplicative constant, this allows the normalisation $\alpha_r = 1$.

We define the **characteristic polynomials** for LMM: $\rho(\zeta) = \sum_{i=0}^r \alpha_i \zeta^i$ and $\sigma(\zeta) = \sum_{i=0}^r \beta_i \zeta^i$

Adams–Bashforth methods

Adams–Bashforth schemes are **explicit** LMM. They can be written letting $\alpha_r = 1$, $\alpha_{r-1} = -1$, $\alpha_j = 0$ for $j \leq r - 2$, $\beta_r = 0$:

$$U_{n+r} = U_{n+r-1} + \Delta t \sum_{j=0}^{r-1} \beta_j f(U_{n+j}, t_{n+j})$$

They amount to approximate the integral in

$$u(t_{n+r}) = u(t_{n+r-1}) + \int_{t_{n+r-1}}^{t_{n+r}} u'(t) dt = u(t_{n+r-1}) + \int_{t_{n+r-1}}^{t_{n+r}} f(u(t), t) dt$$

by a quadrature rule (hence by interpolating $u(t)$ by a polynomial on the $r + 1$ points).

Adams–Bashforth methods: examples

The lowest-order Adams–Bashforth schemes are:

1-step $U_{n+1} = U_n + \Delta t f_n$ (forward Euler)

2-step $U_{n+2} = U_{n+1} + \frac{\Delta t}{2} (-f_n + 3f_{n+1})$

3-step $U_{n+3} = U_{n+2} + \frac{\Delta t}{12} (5f_n - 16f_{n+1} + 23f_{n+2})$

4-step $U_{n+4} = U_{n+3} + \frac{\Delta t}{24} (-9f_n + 37f_{n+1} - 59f_{n+2} + 55f_{n+3})$

where $f_j \equiv f(U_j, t_j)$.

Adams–Moulton methods

Adams–Moulton schemes are **implicit** LMM. They can be written letting $\alpha_r = 1$, $\alpha_{r-1} = -1$, $\alpha_j = 0$ for $j \leq r - 2$, the derivation is similar to that of Adams–Bashforth schemes:

$$U_{n+r} = U_{n+r-1} + \Delta t \sum_{j=0}^r \beta_j f(U_{n+j}, t_{n+j})$$

The lowest-order Adams–Moulton schemes are:

1-step $U_{n+1} = U_n + \frac{\Delta t}{2}(f_n + f_{n+1})$ (Crank–Nicolson)

2-step $U_{n+2} = U_{n+1} + \frac{\Delta t}{12}(-f_n + 8f_{n+1} + 5f_{n+2})$

3-step $U_{n+3} = U_{n+2} + \frac{\Delta t}{24}(f_n - 5f_{n+1} - 19f_{n+2} + 9f_{n+3})$

BDF schemes

Other very interesting LMM methods for CFD are BDF methods for which $\beta_j = 0$ for $j < r$. They are **implicit** methods very well suited for stiff ODEs.

The general scheme has the form: $\sum_{j=0}^r \alpha_j U_{n+j} = \Delta t \beta_r f(U_{n+j}, t_{n+j})$.

The most employed schemes are:

1-step $U_{n+1} - U_n = \Delta t f_{n+1}$ (implicit Euler)

2-step $3U_{n+2} - 4U_{n+1} + U_n = 2\Delta t f_{n+2}$

3-step $11U_{n+3} - 18U_{n+2} + 9U_{n+1} - 2U_n = 6\Delta t f_{n+3}$

4-step $25U_{n+4} - 48U_{n+3} + 36U_{n+2} - 16U_{n+1} + 3U_n = 12\Delta t f_{n+4}$

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Are you satisfied?

We have seen a lot of numerical schemes to integrate systems of ODEs, but:

- An unstable integration scheme is useless.
- Scheme accuracy and stability must be balanced to achieve the desired accuracy at the least cost.

The big questions

Are these methods stable?

How accurate are these methods?

In this section we try to answer these questions.

Formal order of accuracy

The formal order of accuracy of a finite difference scheme can be deduced as follows:

- 1 Starting from the ODE (PDE) approximation obtained by an integration (Finite-Difference) scheme
- 2 Eliminate all nodal values except the central one by Taylor expansion
- 3 Collect the original ODE (PDE) and eliminate it
- 4 The leading powers of Δt (Δx), as $\Delta t \rightarrow 0$ ($\Delta x \rightarrow 0$), in the remaining terms indicate the order of accuracy of the method

A numerical scheme with formal order of accuracy > 0 is said **consistent**.

Formal order of accuracy: example

Let us consider, e.g., the ODE $u'(t) = f(u(t))$ approximated by a 2-step BDF scheme:

$$\frac{3U_n - 4U_{n-1} + U_{n-2}}{2\Delta t} = f_n = U'_n$$

Let us introduce the expansions:

$$U_{n-1} = U_n - \Delta t U'_n + \Delta t^2 U''_n / 2 - \Delta t^3 U'''_n / 6 + o(\Delta t^4)$$

$$U_{n-2} = U_n - 2\Delta t U'_n + 2\Delta t^2 U''_n - 4\Delta t^3 U'''_n / 3 + o(\Delta t^4)$$

By introducing these expansions in the ODE approximation

$$\begin{aligned} & 3U_n/\Delta t - 4U_n/\Delta t + 4U'_n - 2\Delta t U''_n + 2\Delta t^2 U'''_n / 3 \\ & + U_n/\Delta t - 2U'_n + 2\Delta t U''_n - 4\Delta t^2 U'''_n / 3 + o(\Delta t^3) = 2f_n \end{aligned}$$

The residual vanishes as Δt^2 therefore the method is second order.

The concept of 0-stability

The **formal** order of accuracy is not sufficient to estimate the difference between the true solution $u(t)$ and its numerical approximation U_n . The numerical scheme has also to be **stable**.

0-Stability: stability in the limit $\Delta t \rightarrow 0$

A finite-difference scheme is considered stable if, for any time T , there exists a constant C_T independent of Δt such that

$$\|\Phi^n\| \leq C_T \|\Phi^0\| \quad \text{for all } n\Delta t \leq T$$

Beware

0-stability does NOT mean that the scheme is stable for all Δt !

0-stability for LMM

0-stability can be easily predicted for LMM, in fact:

An r -step LMM is 0-stable if the roots of its characteristic polynomial $\rho(\zeta)$ satisfy the following conditions:

$$|\zeta_i| \leq 1 \quad \forall 0 \leq i \leq r.$$

If $|\zeta_i|$ is a multiple root

$$|\zeta_i| < 1.$$

Convergence

A finite difference scheme is said to be **convergent** of order (p, q) if

$$\|\phi(n\Delta t, i\Delta x) - \Phi_i^n\| \leq C(\Delta t)^p(\Delta x)^q$$

for $\Delta t \rightarrow 0$, $\Delta x \rightarrow 0$.

In other words, a convergent scheme is a scheme such that the error goes to zero as the refinement, in space and time, is increased.

A convergent scheme is consistent and 0-stable.

Lax's equivalence theorem

If a finite-difference scheme is

- linear
- stable
- formally accurate of order (p, q)

then it is convergent of order (p, q) .

While it is quite easy to obtain the formal accuracy of a finite difference scheme, it is less obvious to prove its stability.

Two similar theorems apply also to BVP for PDEs and IVP for ODEs.

Absolute stability

As said, a convergent scheme could be unstable for a sufficiently large time step.

The next question is, therefore, **which is the time step limit** to have a stable integration? The answer is related to the behaviour of the integration scheme when applied to the linear equation

$$u' = \lambda u, \lambda \in \mathbb{C}.$$

Absolute stability and A-stability

The region of absolute stability of the scheme is the region in the complex plane $\Delta t \lambda$ where U_n decays as $n \rightarrow \infty$.

A scheme is a-stable if its region of absolute stability includes the half plane $\Re(\Delta t \lambda) \leq 0$.

N.B. A scheme is 0-stable if and only if the region of absolute stability includes the origin.

Absolute stability for LMM

Also in this case there is a simple rule to trace the region of absolute stability when we consider a LMM.

In fact:

Absolute stability

The region of absolute stability for a LMM is the set of points z in the complex plane for which the roots ζ_i of the polynomial $\rho(\zeta) - z\sigma(\zeta)$ satisfy:

$$|\zeta_i| \leq 1 \quad \forall 0 \leq i \leq r.$$

If $|\zeta_i|$ is a multiple root

$$|\zeta_i| < 1.$$

Example: the Crank–Nicolson scheme

Let us consider the Crank-Nicolson scheme. We have

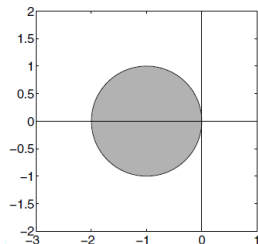
$$\rho(\zeta) = \zeta - 1, \sigma(\zeta) = \frac{\zeta + 1}{2}.$$

then, the polynomial is $\zeta - 1 - z(\zeta + 1)/2$. The method is absolutely stable for $z = \lambda\Delta t$ such that

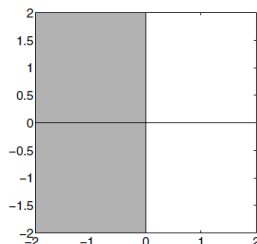
$$|\zeta| = \left| \frac{2+z}{2-z} \right| \leq 1.$$

Which can be easily shown to correspond to the plane $\Re(z) \leq 0$. The method is therefore A-stable.

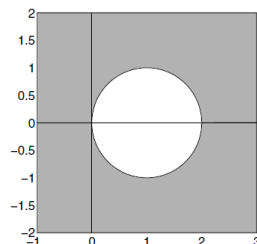
Stability regions for some schemes



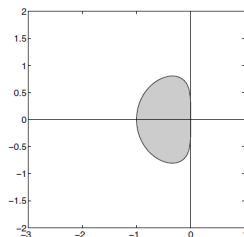
A-B 1



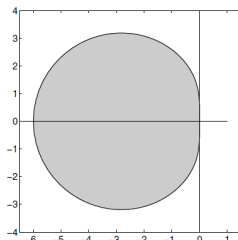
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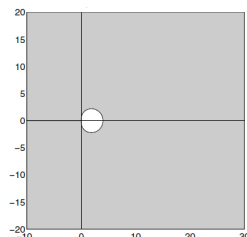
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A-B 2



A-M 2



BDF 2

Systems of ODEs

We do not give any rigorous results, but rules of thumb:

Linear systems: rule of thumb

Writing the system in the form $u' = Au$, where $u(t)$ is the vector unknown, the linearisation scheme is stable if $\Delta t \lambda_i$ belongs to the region of stability of the scheme for all eigenvalues λ_i of the matrix A .

Nonlinear systems: rule of thumb

In this case, we write the linearised system $u' = Ju$ where $J = df/du$ is the Jacobian matrix of the vector function $f(u, t)$. The scheme will be stable if the eigenvalues of the Jacobian matrix satisfy the above condition.

Stiff ODEs

Stiff problem: the solution is smooth and slowly varying (relative to the time interval of the computation), but nearby solution curves are much more rapidly varying, with a short-lived **transient** response that moves the solution back toward a smooth solution.

Stiffness is measured by the ratio:

$$\frac{\max |\lambda_p|}{\min |\lambda_p|}.$$

N.B.1 This ratio does not tell the whole story.

N.B.2 The (unsteady) Stokes equation is stiff.

Integration schemes for stiff problems

The numerical scheme must be **A-stable** to efficiently integrate stiff ODEs.

Theorem (Dahlquist)

Any A-stable LMM is at most second order accurate. The Crank-Nicolson method is the A-stable method with smallest truncation error.

Some A-stable schemes:

- Backward Euler
- Crank–Nicolson
- BDF2 method

One-step methods and L-stability

A useful concept can be introduced for one-step methods: L-stability. The problem is that A-stability assures us that a small perturbation will not grow, but it does not assure that it will decay. This property is assured by L-stability.

We can write a one-step method for $u' = \lambda u$ as: $U_{n+1} = R(z)U_n$, $z = \Delta t\lambda$, then

L-stability for one-step methods

A method is L-stable if it is A-stable and $R(z) \rightarrow 0$ as $z \rightarrow \infty$.

L-stable methods are most suitable for stiff problems.

The Crank-Nicolson method is not L-stable, while the backward Euler method is.

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The case of a discretised PDE

To insure that a numerical scheme will be convergent, we must insure that the method is consistent and stable.

- The first requirement can be checked trivially.
- The second requirement asks that the spectrum of the discretised transition matrix (A in $U_{n+1} = AU_n$) be in the stable region as $\Delta t \rightarrow 0$. This could impose a limitation on Δt as a function of the space-grid refinement. To be more precise, we must ask that

$$\Delta t \lambda_i(\Delta x, \Delta y, \Delta z) \in \text{stability region of the scheme as } \Delta t, \Delta x, \Delta y, \Delta z \rightarrow 0.$$

N.B. This is related to the fact that, in general, the spectrum of A spreads as $\Delta x, \Delta y, \Delta z \rightarrow 0$.

Problem

The problem with convergence analysis is related to the **stability analysis**.

It is not trivial to prove the stability of a method.

Two different approaches can be pursued, with a different degree of difficulty and different generality:

- analyse the stability of the full system, comprehensive of boundary conditions. This approach requires functional analysis.
- analyse the stability of a simplified system, where periodic boundary conditions are applied instead of the correct b.c.: Von Neumann analysis.

Von Neumann analysis

Von Neumann analysis is restricted to constant-coefficient linear problems in unbounded domains, but it can give useful indications on the performance of the numerical schemes.

How does the Von Neumann method work?

- We assume the solution as being a superposition of normal modes $U_n^\xi(x_j) = e^{ijh\xi}$ of different wavenumbers ξ , where $h \equiv \Delta x$.
- We put this assumption in the discretised PDE and obtain a relation $U_{n+1}^\xi(x_j) = g(\xi)e^{ijh\xi}$
- We ask that $|g(\xi)| \leq 1$ for stability

Since the problem is linear, normal modes evolve independently from each other.

Example: diffusion equation and the Forward-Euler scheme

Let us analyse the stability of a forward-Euler+centred difference scheme for the equation

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2} \rightarrow U_{n+1,j} = U_{n,j} + \frac{\nu \Delta t}{\Delta x^2} (U_{n,j+1} - 2U_{n,j} + U_{n,j-1})$$

Let us introduce $U_{n,j} = e^{ijh\xi}$, $U_{n,j+1} = e^{i(j+1)h\xi}$ and $U_{n,j-1} = e^{i(j-1)h\xi}$ in the previous equation, we obtain after some algebraic manipulations

$$U_{n+1,j} = \left(1 + 2\frac{\nu \Delta t}{\Delta x^2} (\cos(h\xi) - 1)\right) e^{ijh\xi} \Rightarrow g(\xi) = 1 + 2\frac{\nu \Delta t}{\Delta x^2} (\cos(h\xi) - 1)$$

which gives the limit $\Delta t \leq \frac{\Delta x^2}{2\nu}$.

Example: a pure transport equation

Let us analyse another PDE, also very interesting for fluid mechanics. We use a similar discretization (FE+centred differences)

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \rightarrow U_{n+1,j} = U_{n,j} - \frac{a \Delta t}{2 \Delta x} (U_{n,j+1} - U_{n,j-1})$$

By the same approach adopted in the previous case, we obtain

$$g(\xi) = 1 - i \frac{a \Delta t}{\Delta x} \sin(h\xi), |g(\xi)| \geq 1!$$

The method is unconditionally unstable!

We can cure this by using a one sided (upwind) approximation of the spatial derivative, in this case we obtain a stability limit

$$0 \leq \frac{a \Delta t}{\Delta x} \leq 1.$$

The Courant-Friedrichs-Lewy condition

A condition such as the one found for the previous problem is called Courant-Friedrichs-Lewy condition.

The **Courant-Friedrichs-Lewy condition** (CFL) has the form

$$\left| \frac{a \Delta t}{\Delta x} \right| \leq C.$$

It applies frequently in fluid mechanics, especially when direct numerical simulations of the Navier–Stokes equations are carried out by taking into account diffusion implicitly and convection explicitly.

Conclusion

We can exploit the previous results to draw some conclusions on the time discretization of the Navier–Stokes equations:

- since the generalised Stokes equation is stiff (the stability limit depends on Δx^2) this operator is generally discretised implicitly using either a Crank-Nicolson or a BDF2 scheme. Indeed, this part of the operator does not depend on time and can be factored once and for all at the beginning of the computation.
- the convection part, which gives a less stringent limit (CFL condition), is discretized explicitly, owing to its nonlinearity that would require the solution of a nonlinear equation for each time step if discretised implicitly.

Exercise

Try at least three different time-advancement (both implicit and explicit) schemes to solve:

$$\frac{\partial T}{\partial t} - k \nabla^2 T = f(x, t).$$

with Dirichlet or Neumann boundary conditions and a manufactured solution.

- Observe the behaviour of the numerical solution as a function of the time step.
- For a stable scheme, plot the error as a function of the time step for different spatial discretisations.
- Check that the predicted convergence rate is actually observed using a manufactured solution.
- Comment the results.

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Some references

- R.J. Leveque, *Finite Difference Methods for Ordinary and Partial Differential Equations*, SIAM, 2007.